From Molecule to Megabyte



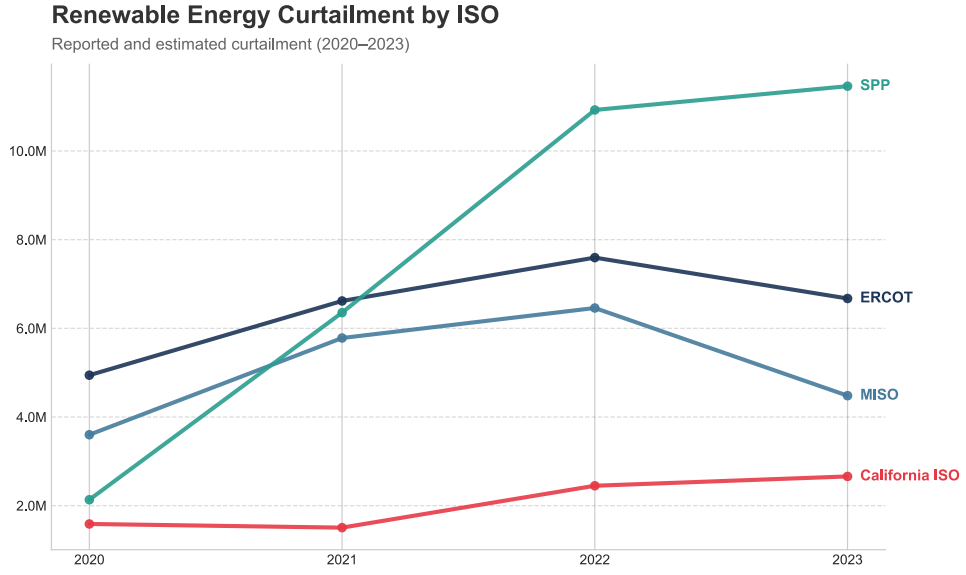
Sierra Solutions

Energython 2026 | International Champion

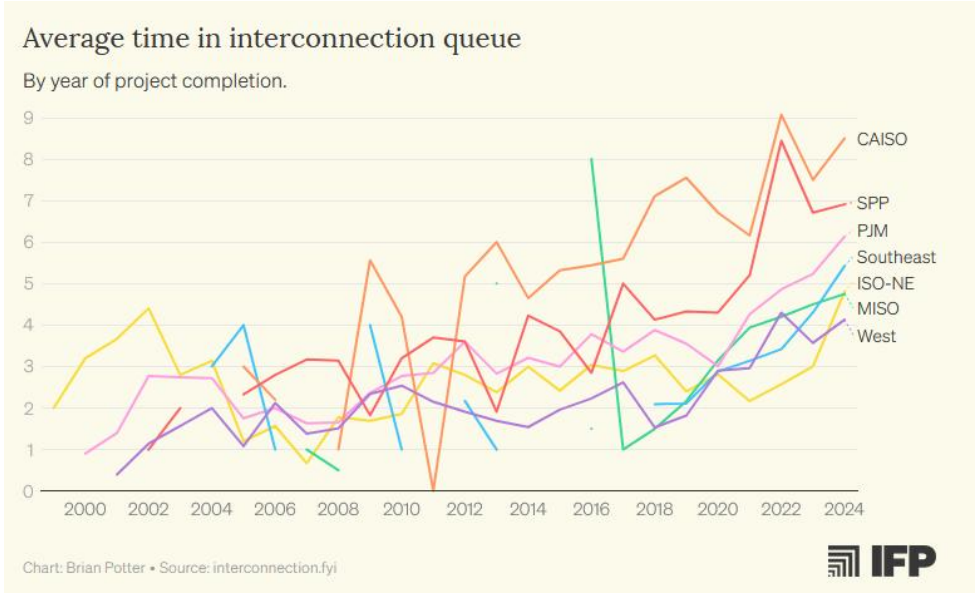
AI is Growing. Grid is *Not*.

The physical bottleneck of the digital revolution.

- AI demand is doubling every six months.
- Interconnection queues for grid power now exceed **60 months**.
- Curtailments increased **from 40% up to 500%** in different ISOs.
- **Grid curtailments** & **transmission backlogs** are a **death sentence for high-uptime AI inference**.



spglobal.com

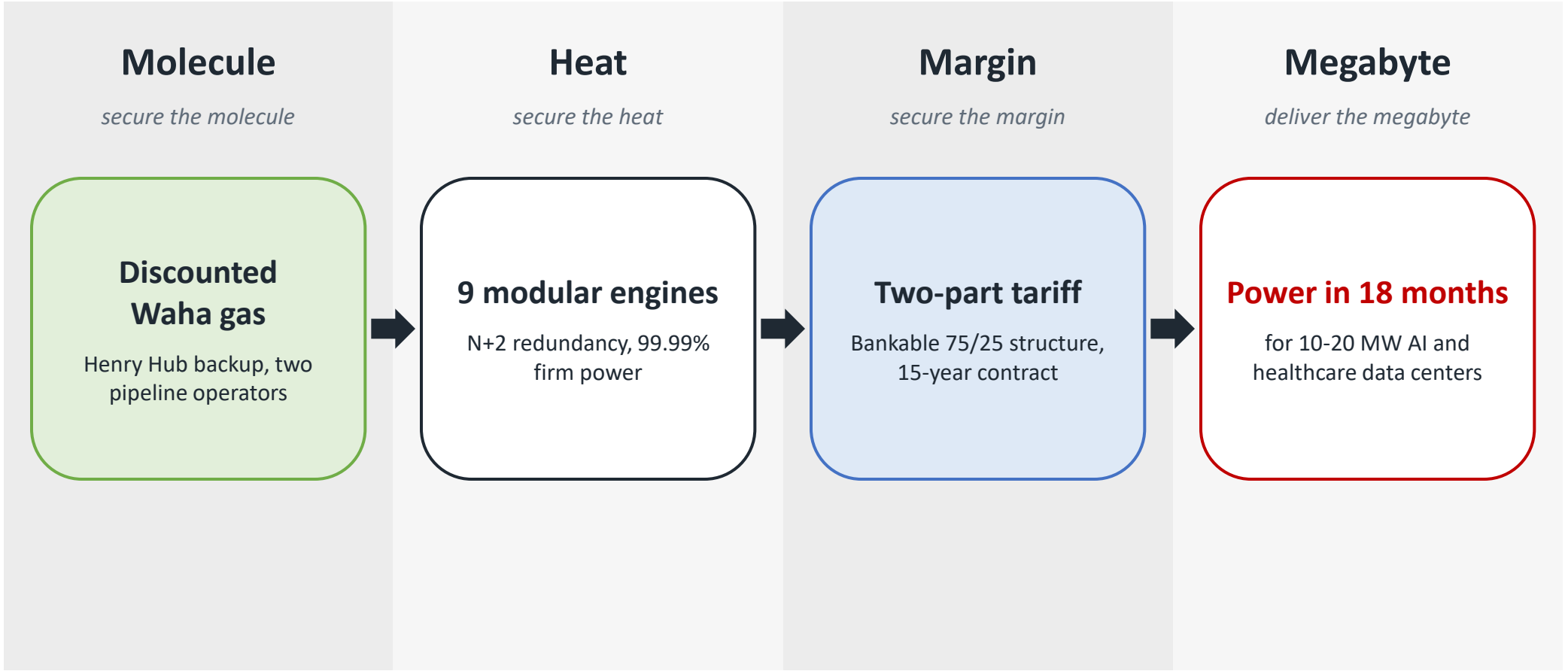


Interconnection.fyi

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From Molecule to Megabyte

A high-reliability behind-the-meter power solution by Sierra Solutions.

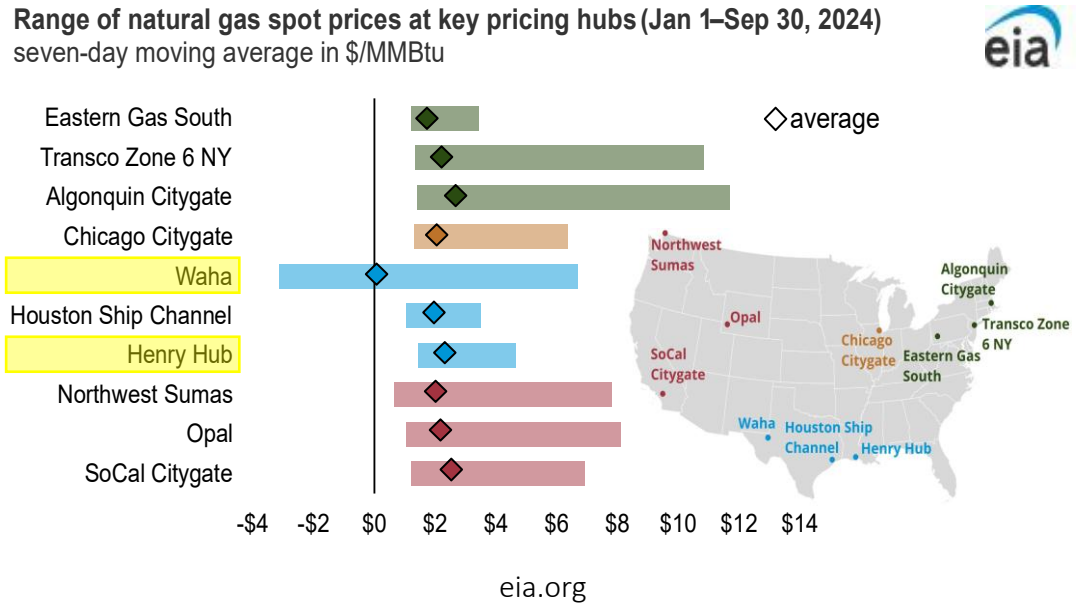


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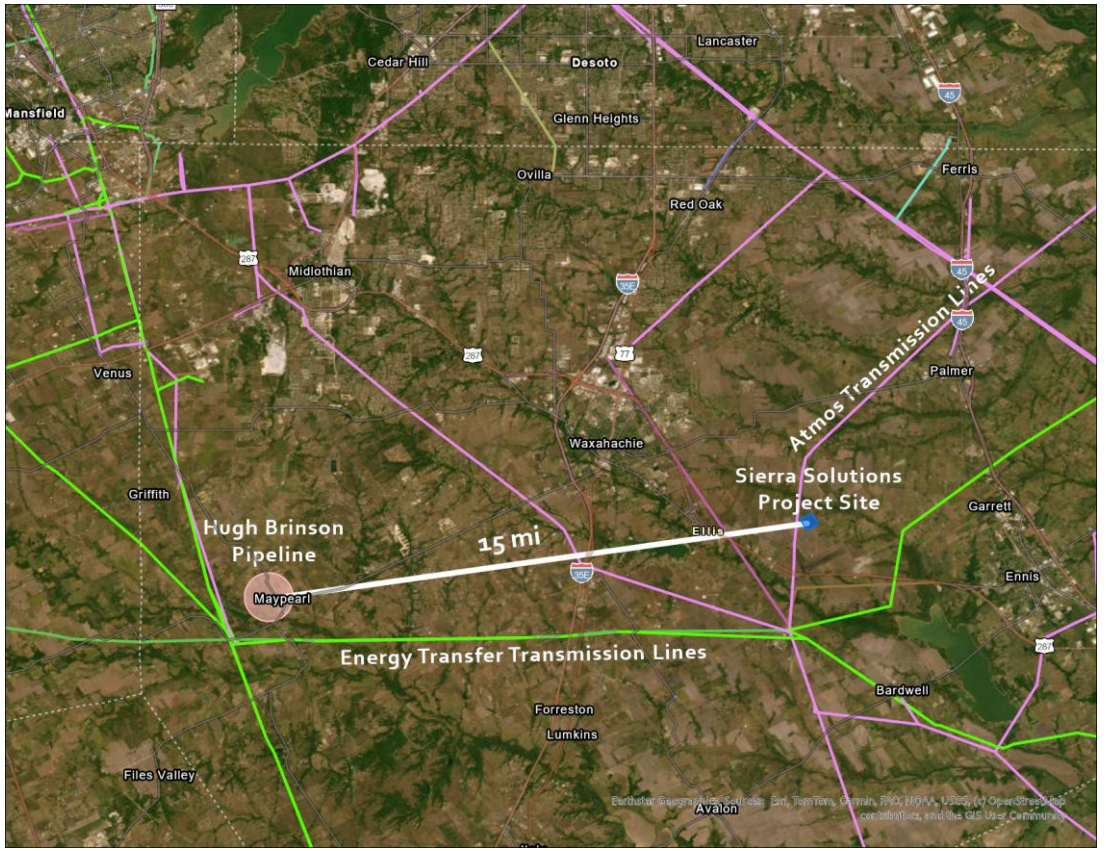
Why Dallas

High demand meeting physical gas optionality.

- Primary: Waha Hub (Cheapest in the US).
- Safety Net: Henry Hub (Most liquid in the US).
- Double-hub connectivity **hedges supply risk**.



eia.org



Sierra Solutions

Technology Refinement

Modular engines outperforming turbines for high-reliability 15MW loads.

		N Configuration		N+1 Configuration		N+2 Configuration		2N Configuration	
Name	Price ($M/unit)	Units (N)	Total CapEx ($M)	Units (N+1)	Total CapEx ($M)	Units (N+2)	Total CapEx ($M)	Units (2N)	Total CapEx ($M)
Titan 130	10.925	2	21.85	3	**32.78**	4	43.70	4	43.70
W31SG	17.655	2	35.31	3	**52.97**	4	70.62	4	70.62
G3520H	1.725	7	12.08	8	13.80	**9**	**15.53**	14	24.15

Titan 130: 4 units for N+2 ($43.7M); one trip drops 76% of load; loses output in Texas heat.

Wartsila 31SG: 4 units for N+2 ($70.6M); 4.5x our cost for the same redundancy.

Cat G3520H: 9 units for N+2 ($15.5M); one trip drops only 14% of load. Selected.



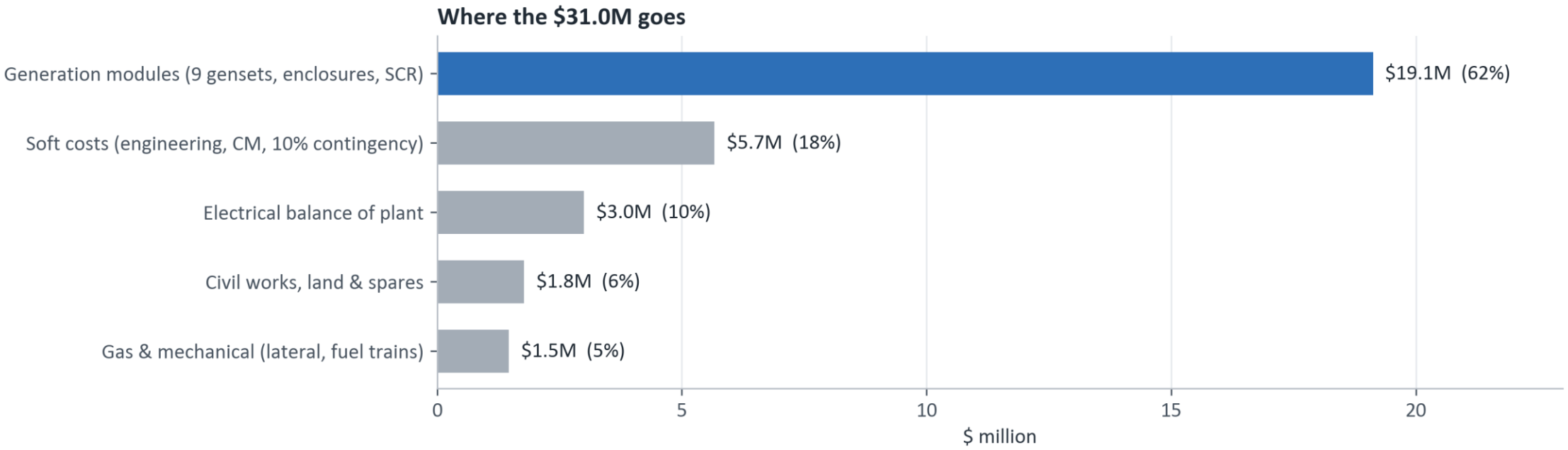
westernstatescat.com

- **9 modular** CAT G3520H gensets in N+2: 15.75 MW firm / 20.25 MW installed, radiator-cooled (no water).
- **N+2** delivers 99.99% firm availability; N+1 would give only 99.73% (~24 h/yr exposed).
- **Honest physics**: CAT's 8,114 Btu/kWh (LHV) = 8,990 Btu/kWh on the HHV basis we buy gas on, +0.5%/yr degradation.

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Project CAPEX

62% of the budget is generation equipment; cost per kW sits mid-range.



- Includes $2.8M P50 contingency and $1.5M permitting/engineering.
- $1,532/kW installed ($1,970/kW firm), within NREL's $900-$2,600/kW range [1].

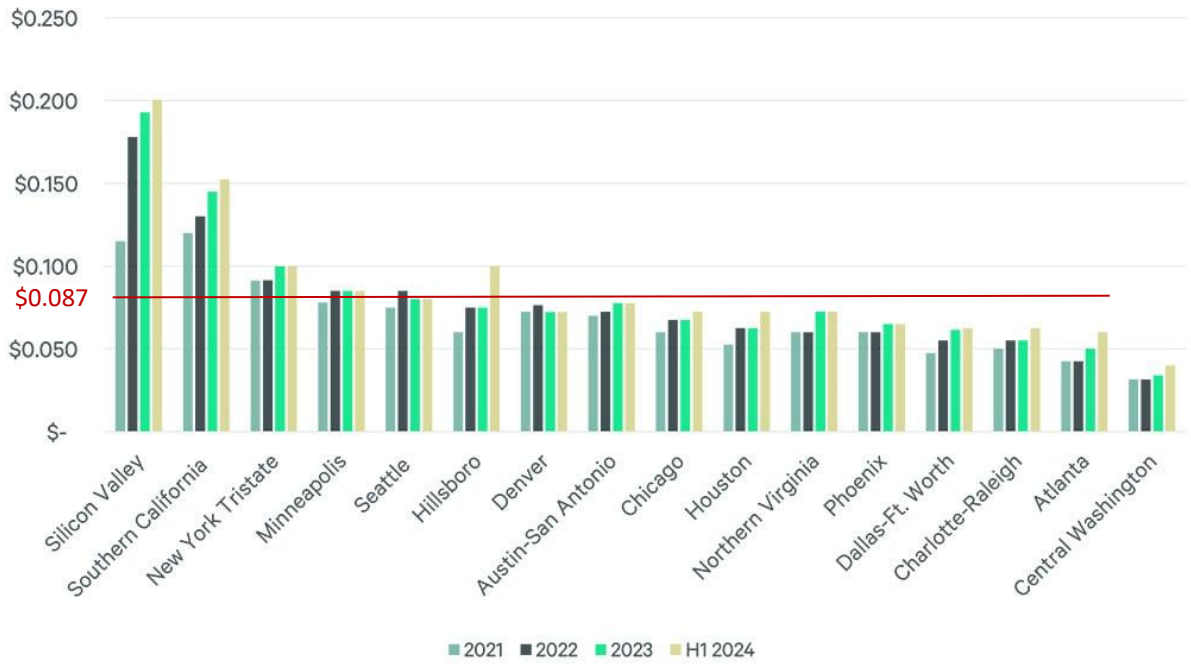
[1] nrel.gov

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Reliability is Affordable

A Two-Part Tariff designed to protect debt service and equity upside.

- Capacity Rate: **$22.50 / kW-mo.**
  - (Reservation fee on 15.75 MW firm, 99.99% uptime).
- Energy Rate: **$0.055 / kWh.**
- 8.7 cents/kWh blended rate—highly competitive for on-site prime power.
  - 40-55% cheaper than California grids.
  - ~2.2 cents/kWh above the Texas industrial average (6.55 cents, EIA 2025).
  - That premium (~$2.9M/yr) buys a 3.5-year head start (~$100M of revenue) and no curtailment risk.



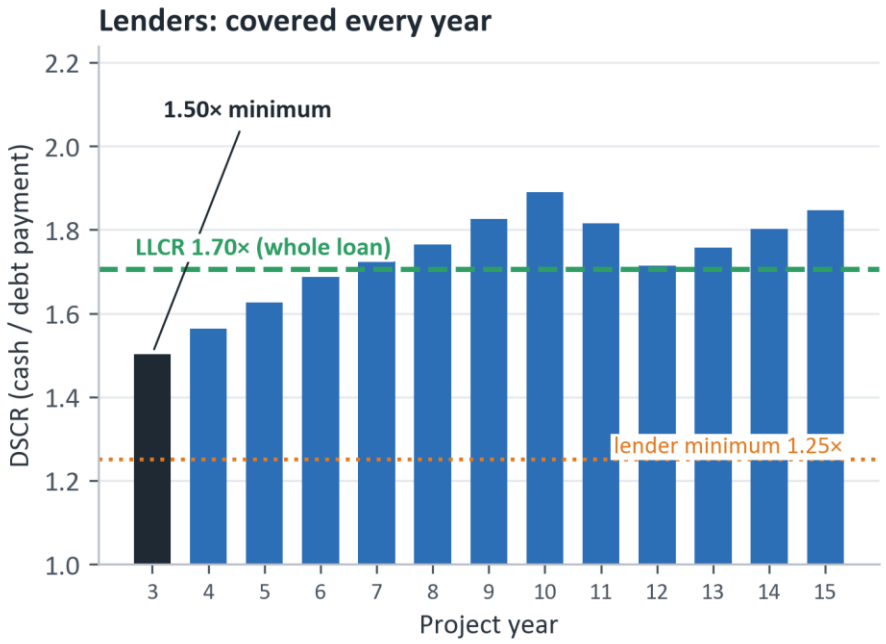
cbre.com

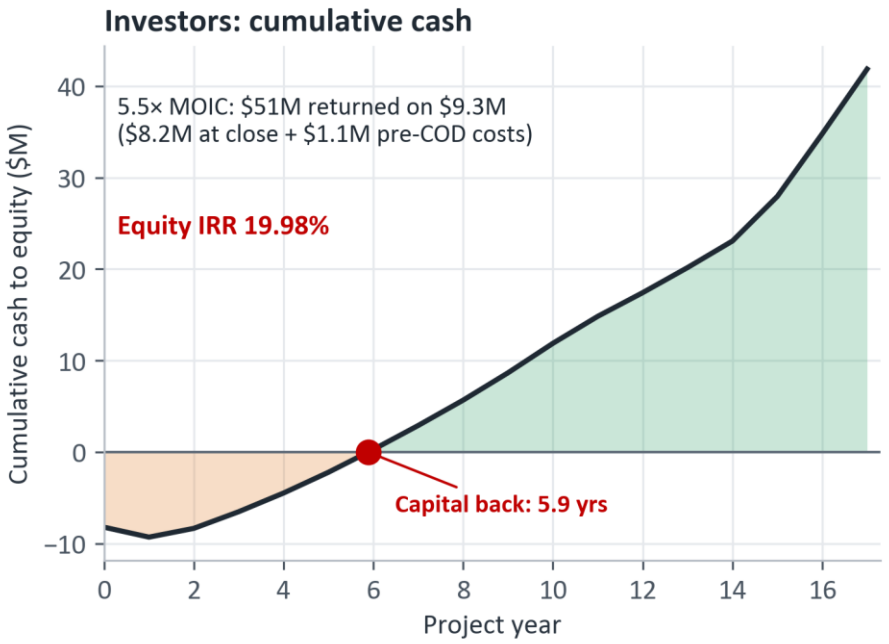
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Bankable Margins

Lenders covered 1.50× every year; investors earn 19.98%.

- **Lenders:** capacity payments alone cover 125% of debt service; DSCR never below 1.50×, LLCR 1.70× (minimums 1.25x / 1.40x).
- **Investors:** **19.98% equity IRR, 5.5× MOIC**, capital back in 5.9 yrs.
- **Structure:** 75% senior debt / 25% equity; 13-year loan ends 2 years before the 15-year contract.



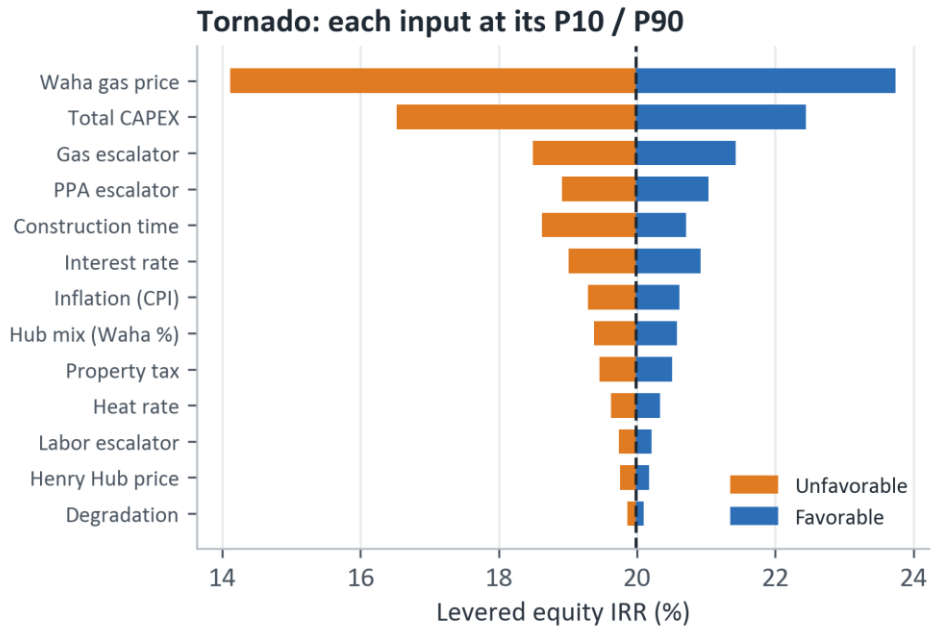


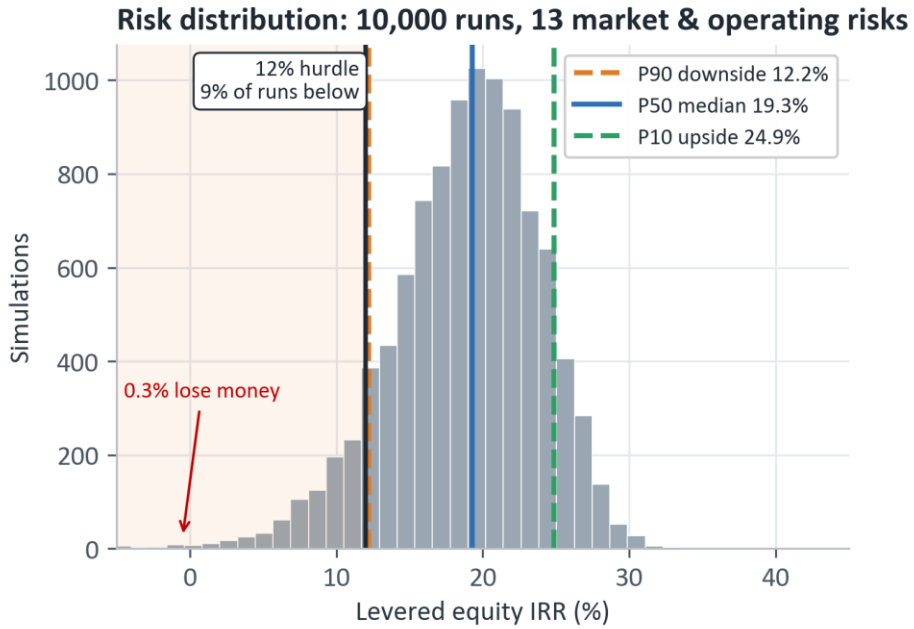
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Value Driver Analysis

Stress-testing the 19.98% IRR: 10,000 runs across 13 market and operating risks.

- **13 risks**, each modeled from real data: 7 years of gas-hub prices, cost and schedule overrun tails.
- **P90 of 12.2%**: 9 in 10 outcomes clear the 12% hurdle.
- Median 19.3%; **upside (P10) 24.9%** if the Waha discount persists.



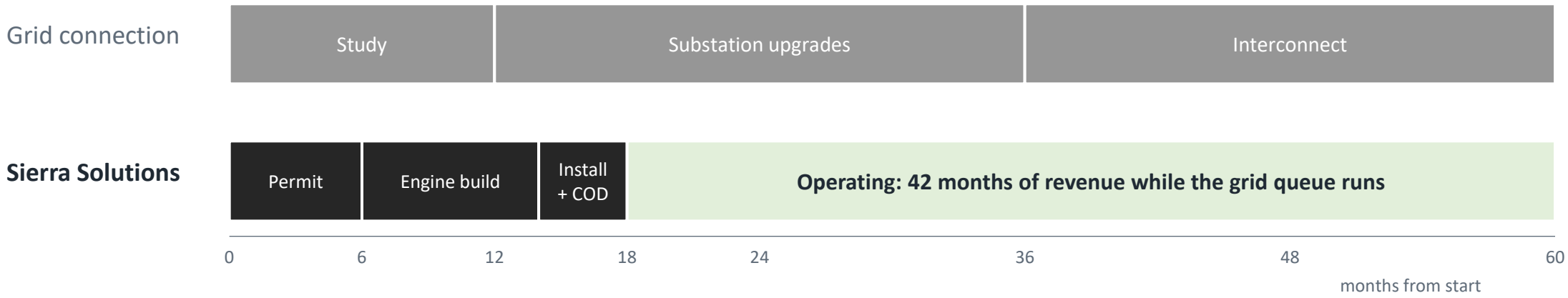


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Winning the Race to Data

18 months to power vs. 60 for a grid connection, drawn to scale.

- Sierra Solutions bypasses the interconnection queue (4+ years in ERCOT, 5+ years average).
- Built for 10-20 MW AI-inference, GPU-cloud and healthcare data centers that cannot wait for the grid.



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Invest in Project Dallas

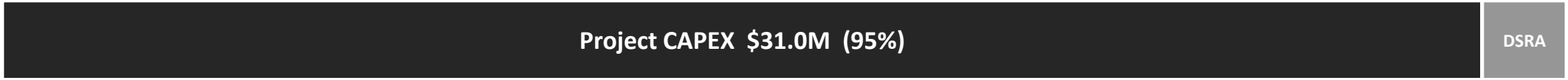
Seeking $8.2M in equity to build Project Dallas.

19.98%	5.5×	5.9 yrs	$6.92M
equity IRR target 12-25%	MOIC target at least 1.5x	payback from financial close	NPV at a 12% hurdle

Sources: $32.8M



Uses: $32.8M



DSRA: $1.81M debt-service reserve (6 months of payments), released at loan maturity.

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Thank you



Sierra
Solutions

Q&A Slides

Sierra Solutions

Energython 2026 | International Champion

Q&A Navigator

Click a topic to jump; every Q&A slide links back here.

1. Annual cash flow
2. OPEX breakdown
3. Reading the cash-flow chart
4. DSCR profile and tax cliff
5. Loan tenor: 13 vs. 15 years
6. Capital structure: 75% vs. 65%
7. Load factor sensitivity
8. Tax depreciation sensitivity
9. Exit options
10. Customer profile
11. Customer examples in DFW
12. Sierra Solutions business model
13. Project schedule (18 months)
14. Phase 2 expansion
15. Delivery delay vs. returns
16. Scaling beyond 31.5 MW
17. Gas price basis (Waha history)
18. Gas-price resilience
19. Gas supply path
20. Fuel security
21. Is 8.7 cents competitive?
22. Pricing flexibility
23. Monte Carlo inputs
24. Risk metrics: NPV S-curve
25. Risk register
26. Offtaker credit
27. CAPEX line-item detail
28. Heat rate basis: LHV vs. HHV
29. N+2 reliability math
30. Technology selection detail
31. Competitive technology analysis
32. Emissions, water and noise
33. Key agreements
34. Data sources: financial
35. Data sources: technical
36. Engine room
37. Model refinement

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Financial Model: Annual Cash Flow

Finals model: 75% debt, 13-yr tenor, HHV heat rate, Waha $1.15, tax-loss carryforward.

- Escalation: labor 3%, fuel 3%, CPI 2.5%. MACRS 5-year from placed-in-service, losses carried forward (80% limit). DSRA = 6 months of debt service, released at loan maturity.

Year	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Date	1-Jan-27	1-Jan-28	1-Jan-29	1-Jan-30	1-Jan-31	1-Jan-32	1-Jan-33	1-Jan-34	1-Jan-35	1-Jan-36	1-Jan-37	1-Jan-38	1-Jan-39	1-Jan-40	1-Jan-41	1-Jan-42	1-Jan-43	1-Jan-44
Energy Rev.	$0.00 M	$0.00 M	$3.85 M	$7.84 M	$8.08 M	$8.32 M	$8.57 M	$8.83 M	$9.09 M	$9.37 M	$9.65 M	$9.94 M	$10.23 M	$10.54 M	$10.86 M	$11.18 M	$11.52 M	$11.86 M
Capacity Rev.	$0.00 M	$0.00 M	$2.23 M	$4.52 M	$4.64 M	$4.75 M	$4.87 M	$4.99 M	$5.12 M	$5.25 M	$5.38 M	$5.51 M	$5.65 M	$5.79 M	$5.93 M	$6.08 M	$6.24 M	$6.39 M
Total Rev.	$0.00 M	$0.00 M	$6.08 M	$12.37 M	$12.72 M	$13.07 M	$13.44 M	$13.82 M	$14.21 M	$14.61 M	$15.02 M	$15.45 M	$15.88 M	$16.33 M	$16.79 M	$17.27 M	$17.75 M	$18.26 M
Total Fuel Cost	$0.00 M	$0.00 M	$1.18 M	$2.40 M	$2.46 M	$2.53 M	$2.60 M	$2.66 M	$2.74 M	$2.81 M	$2.89 M	$2.96 M	$3.05 M	$3.13 M	$3.22 M	$3.31 M	$3.40 M	$3.50 M
Variable O&M	$0.00 M	$0.00 M	$1.22 M	$2.47 M	$2.53 M	$2.59 M	$2.66 M	$2.72 M	$2.79 M	$2.86 M	$2.93 M	$3.00 M	$3.08 M	$3.16 M	$3.23 M	$3.32 M	$3.40 M	$3.48 M
Staffing	$0.00 M	$0.00 M	$0.49 M	$0.99 M	$1.02 M	$1.05 M	$1.09 M	$1.12 M	$1.15 M	$1.19 M	$1.22 M	$1.26 M	$1.30 M	$1.34 M	$1.38 M	$1.42 M	$1.46 M	$1.50 M
Ins. & Admin	$0.00 M	$0.31 M	$0.32 M	$0.33 M	$0.34 M	$0.34 M	$0.35 M	$0.36 M	$0.37 M	$0.38 M	$0.39 M	$0.40 M	$0.41 M	$0.42 M	$0.43 M	$0.44 M	$0.45 M	$0.46 M
Property Tax	$0.00 M	$0.78 M	$0.78 M	$0.74 M	$0.70 M	$0.67 M	$0.63 M	$0.60 M	$0.57 M	$0.54 M	$0.51 M	$0.49 M	$0.46 M	$0.44 M	$0.42 M	$0.40 M	$0.38 M	$0.36 M
Total Opex	$0.00 M	$1.08 M	$3.98 M	$6.92 M	$7.05 M	$7.18 M	$7.32 M	$7.47 M	$7.62 M	$7.78 M	$7.94 M	$8.12 M	$8.30 M	$8.48 M	$8.68 M	$8.88 M	$9.09 M	$9.31 M
Beginning Bal.	$0.00 M	$24.63 M	$26.72 M	$27.85 M	$26.60 M	$25.24 M	$23.76 M	$22.16 M	$20.42 M	$18.54 M	$16.49 M	$14.27 M	$11.86 M	$9.25 M	$6.41 M	$3.34 M	$0.00 M	$0.00 M
Interest	$0.00 M	$2.09 M	$2.27 M	$2.37 M	$2.26 M	$2.15 M	$2.02 M	$1.88 M	$1.74 M	$1.58 M	$1.40 M	$1.21 M	$1.01 M	$0.79 M	$0.55 M	$0.28 M	$0.00 M	$0.00 M
Cap Int (IDC)	$0.00 M	$2.09 M	$1.13 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M
Principal PMT	$0.00 M	$0.00 M	$0.00 M	$1.25 M	$1.36 M	$1.48 M	$1.60 M	$1.74 M	$1.89 M	$2.05 M	$2.22 M	$2.41 M	$2.61 M	$2.84 M	$3.08 M	$3.34 M	$0.00 M	$0.00 M
End Balance	$24.63 M	$26.72 M	$27.85 M	$26.60 M	$25.24 M	$23.76 M	$22.16 M	$20.42 M	$18.54 M	$16.49 M	$14.27 M	$11.86 M	$9.25 M	$6.41 M	$3.34 M	$0.00 M	$0.00 M	$0.00 M
EBITDA	$0.00 M	$-1.08 M	$2.10 M	$5.44 M	$5.66 M	$5.89 M	$6.12 M	$6.35 M	$6.59 M	$6.83 M	$7.08 M	$7.33 M	$7.59 M	$7.85 M	$8.12 M	$8.39 M	$8.67 M	$8.95 M
MACRS	$0.00 M	$0.00 M	$6.65 M	$10.64 M	$6.38 M	$3.83 M	$3.83 M	$1.92 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M
Taxable Income	$0.00 M	$-1.08 M	$-5.69 M	$-7.57 M	$-2.98 M	$-0.08 M	$0.27 M	$2.56 M	$4.86 M	$5.26 M	$5.68 M	$6.12 M	$6.58 M	$7.06 M	$7.57 M	$8.10 M	$8.67 M	$8.95 M
Income Tax	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.01 M	$0.11 M	$0.20 M	$0.22 M	$0.24 M	$0.76 M	$1.38 M	$1.48 M	$1.59 M	$1.70 M	$1.82 M	$1.88 M
Debt Service	$0.00 M	$0.00 M	$1.14 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$3.62 M	$0.00 M	$0.00 M
LFCF	$-8.21 M	$-1.08 M	$0.96 M	$1.82 M	$2.04 M	$2.27 M	$2.49 M	$2.63 M	$2.77 M	$2.99 M	$3.22 M	$2.95 M	$2.58 M	$2.74 M	$2.90 M	$4.87 M	$6.85 M	$7.07 M
DSCR				1.50	1.56	1.63	1.69	1.73	1.76	1.83	1.89	1.82	1.71	1.76	1.80	1.85		

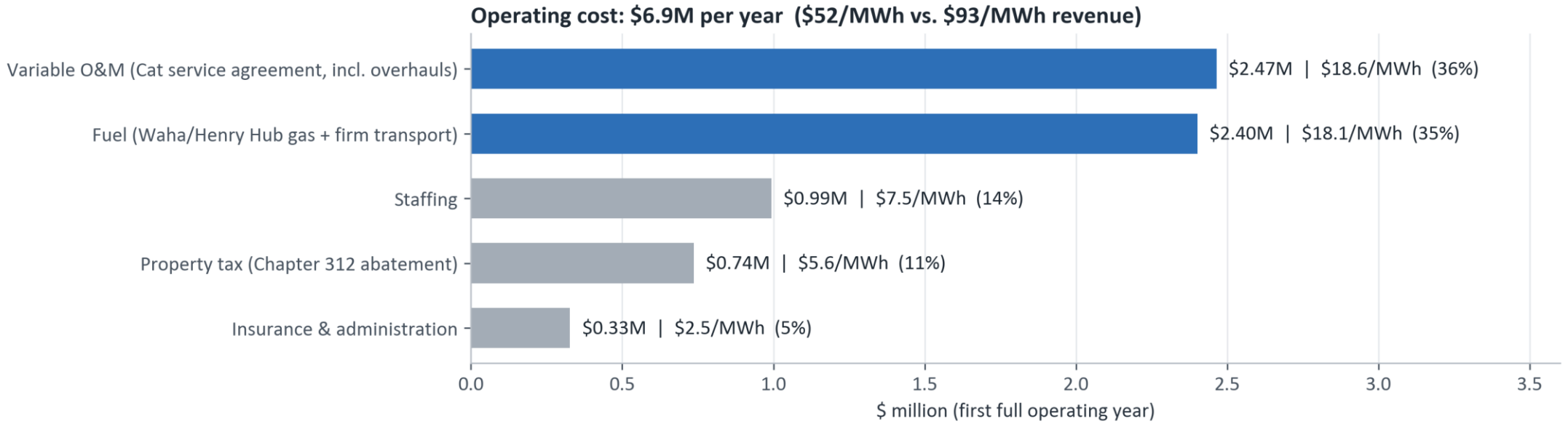
MACRS Year	Rate (%)
1	20
2	32
3	19.2
4	11.52
5	11.52
6	5.76
+7	0

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Operating Costs (OPEX)

Fuel and maintenance are 71% of OPEX; every cost line escalates over 15 years.



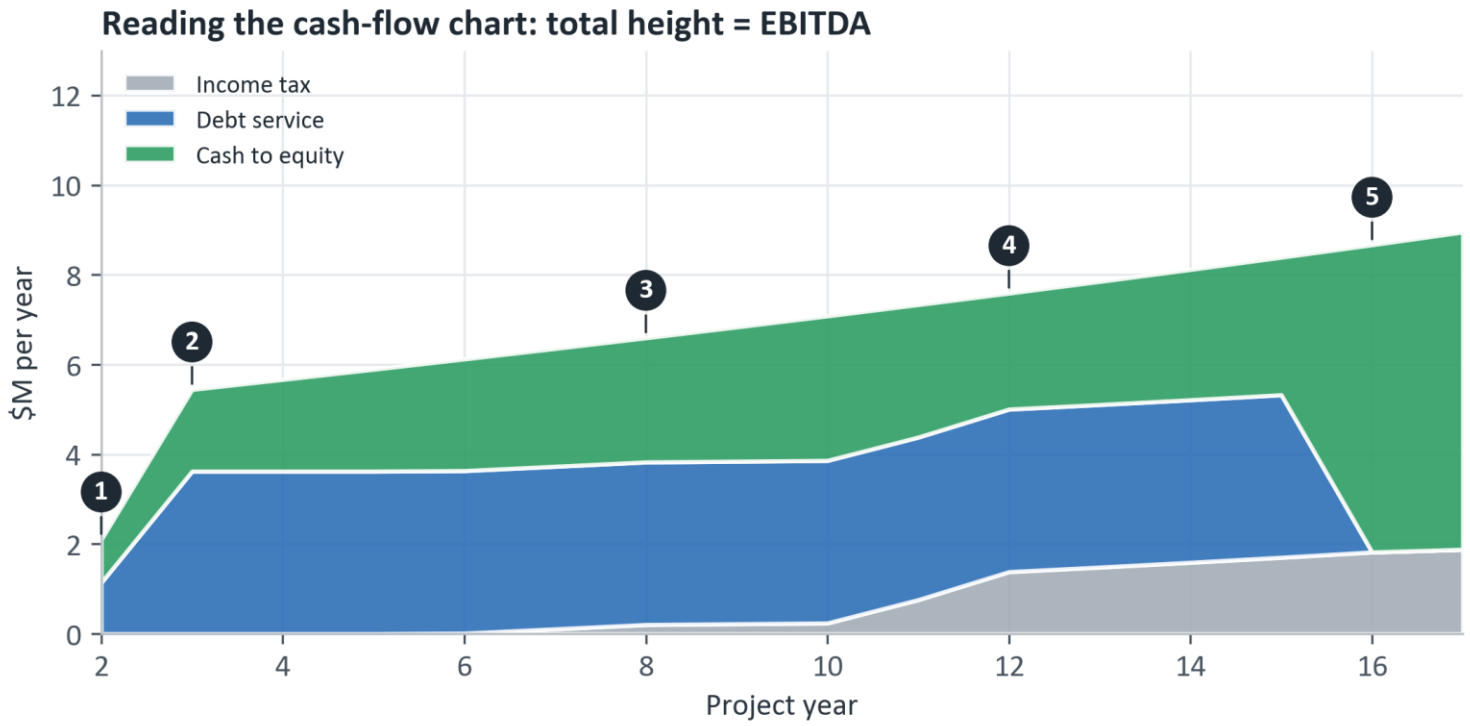
- Fuel: 8,990 Btu/kWh (HHV) x ($1.29 blended Waha/Henry Hub + $0.60 firm transport), +3%/yr gas escalation and 0.5%/yr engine wear.
- Variable O&M $17.50/MWh: Cat service agreement covering overhauls (+2.5%/yr); staffing $0.92M base (+3%/yr).
- OPEX grows from $6.9M to ~$9.3M by year 17, while contracted revenue escalates alongside it.

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Reading the Cash-Flow Chart

Why the bands step up and down.



(1) Year 2: COD mid-year, half a year of revenue

(2) Year 3: first full year, level debt service starts

(3) Years 6-10: loss carryforward shields most income tax

(4) Year 12: tax-loss bank exhausted, full 21% tax

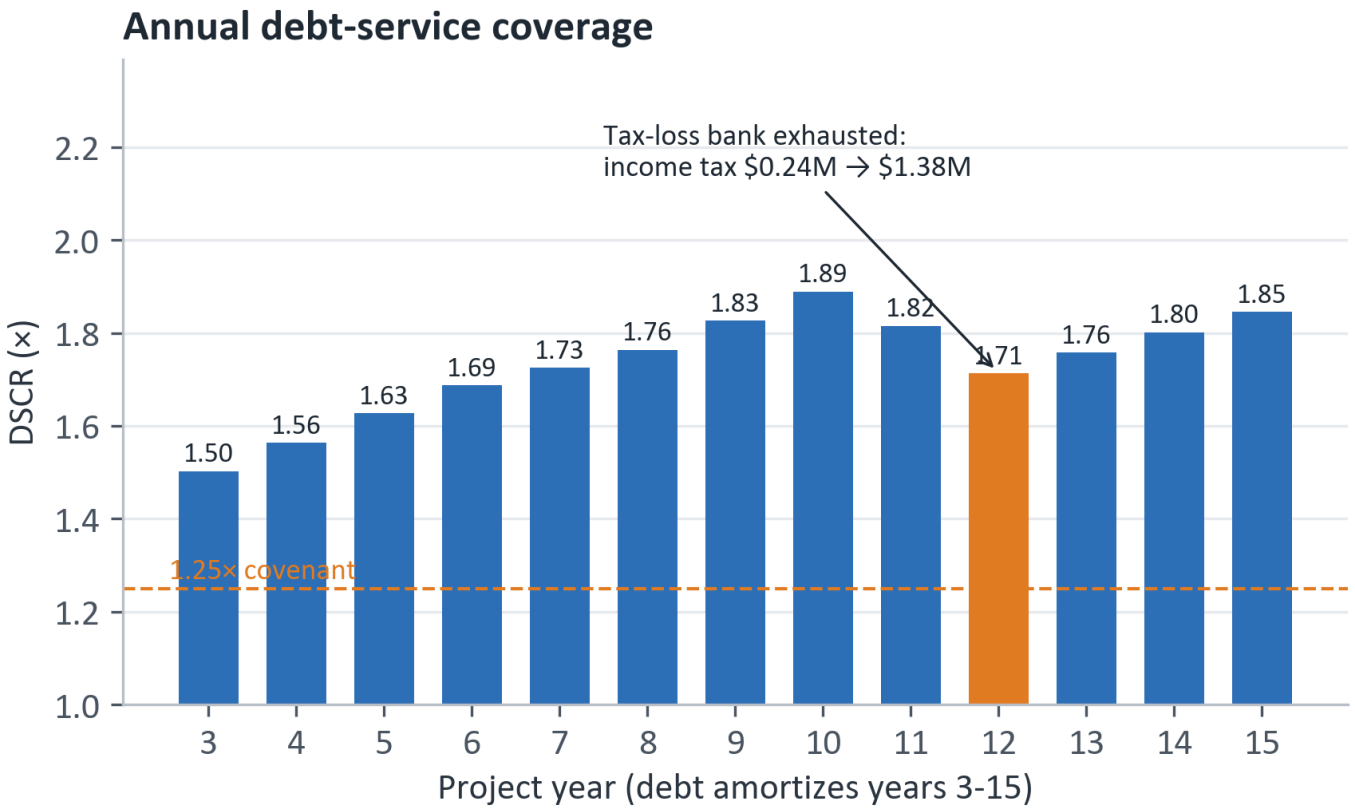
(5) Year 16: loan repaid (2-yr tail); all cash to equity

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DSCR Profile and the Tax Cliff

Coverage never drops below 1.50× in the base case.



- Lowest DSCR is year 3: EBITDA is smallest while the level loan payment starts.
- Coverage then rises with escalating revenue against a flat payment.
- Year 11: the tax-loss bank runs out; year 12 pays full 21% tax ($0.24M to $1.38M), so DSCR steps from 1.89× to 1.71×.
- The dip is expected, and still far above the 1.25x covenant.

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Loan Tenor: 13 vs. 15 Years

Lenders lend against contracted cash flow, and keep a cushion.

Metric	13-yr loan (base)	15-yr loan
Tail inside the 15-yr ESA	2 years	0 years
Equity IRR	19.98%	21.10%
NPV @ 12%	$6.92M	$7.43M
Min DSCR / LLCR	1.50× / 1.70×	1.63× / 1.88×
PLCR (project life)	1.86×	1.88×
MOIC / payback	5.5× / 5.9 yrs	5.3× / 5.5 yrs

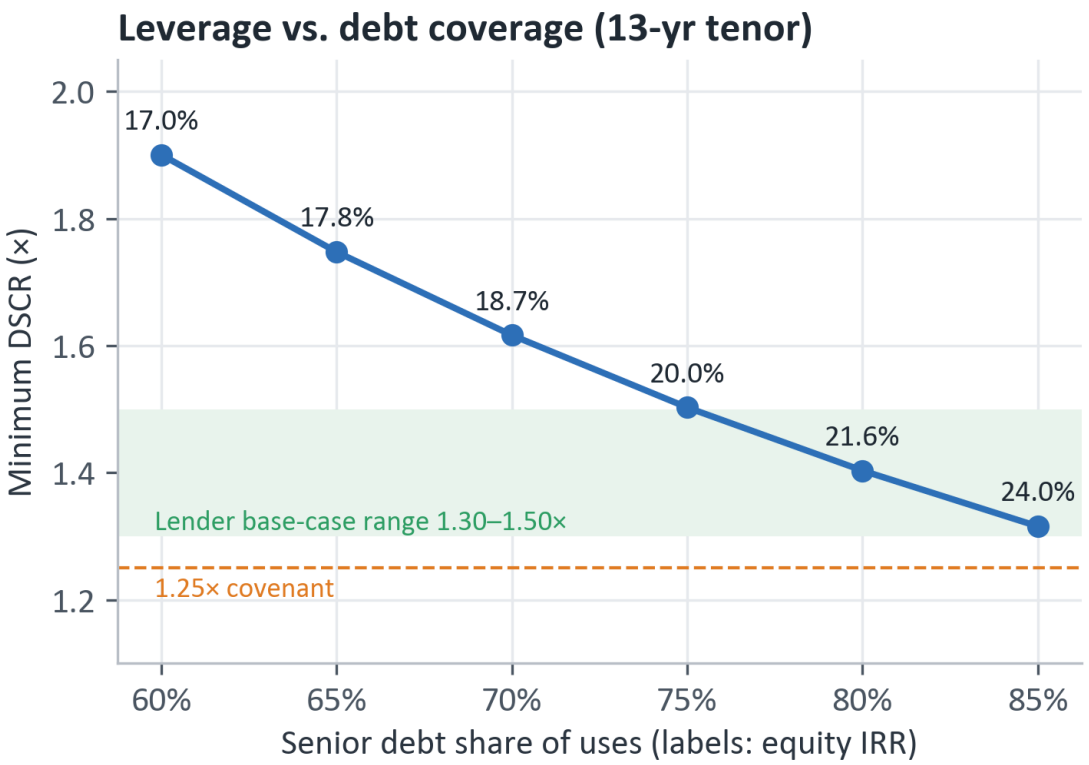
- A tail (contract term minus loan term) gives lenders 1-3 years of contracted cash to restructure if the plant underperforms.
- A loan as long as the contract has zero cushion: lenders would price it higher or demand recontracting evidence.
- The 15-yr loan is worth ~1 pt of IRR to us; we give it up for bankability.
- Engines run 20+ years with overhauls: years beyond the contract and residual value are upside we do not count.

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Capital Structure: 75% vs. 65% Debt

Debt sized to lender coverage; 65% kept as the growth-stage-tenant case.



Metric	75% (base)	65% (conservative)
Equity check	$8.2M	$11.4M
Equity IRR	19.98%	17.76%
NPV @ 12%	$6.92M	$6.20M
MOIC	5.5×	4.6×
Min DSCR / LLCR	1.50× / 1.70×	1.75× / 1.97×
Debt / total cap	77%	68%
P90 / P50 / P10 IRR	12.2% / 19.3% / 24.9%	11.4% / 17.2% / 21.5%
Expected NPV	$6.03M	$5.25M
P(min DSCR < 1.0x)	1.7%	0.5%

75% raises every percentile, including the downside. Investment-grade anchor tenant: 75% at 1.50×. Growth-stage AI tenant: 65% at 1.75×.

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Customer Load Factor Sensitivity

Take-or-pay capacity payments protect the lenders if the data center draws less.

Customer load factor	Equity IRR	Min DSCR	Effective cents/kWh
96%	19.98%	1.50×	8.7¢
90%	18.78%	1.45×	8.9¢
85%	17.75%	1.41×	9.1¢
80%	16.70%	1.37×	9.4¢
70%	14.55%	1.28×	9.9¢

- The capacity charge is 36% of revenue and is paid regardless of load.
- Base case 96%: AI inference runs near flat around the clock.
- At 80% load the project still earns 16.70% with 1.37× DSCR.
- Lower load raises the client's effective cents/kWh, a reason to sign at high utilization.

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Tax Depreciation Sensitivity

Equity returns are robust to the depreciation class.

Depreciation case	Equity IRR	NPV @ 12%	Min DSCR
MACRS 5-year (modeled)	19.98%	$6.92M	1.50×
MACRS 15-year (if IRS class 49.15 applies)	19.38%	$6.48M	1.50×
100% bonus depreciation	19.64%	$6.72M	1.47×

- Modeled: MACRS 5-year (IRS Pub. 946). If the IRS assigns a 15-year class, IRR is 19.38% instead of 19.98%.
- 100% bonus depreciation does not help a standalone project: losses pile up faster than they can be used (80% carryforward limit).
- That value could be captured with a tax-equity partner; it is upside we do not count.

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Exit Options (Illustrative)

How investors get their money back.

- Hold to maturity: 5.5× MOIC, equity payback in 5.9 yrs, contracted yield for 15 years.
- Sell after de-risking: at the end of year 8, at an illustrative 7.0x forward EBITDA (EV $47.8M), equity proceeds of $31.1M give a 27.0% IRR and 5.0× MOIC.
- Refinance after COD: tighter spreads on an operating asset return capital early.
- The exit multiple is illustrative and not part of the base case.

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Customer Profile

Who buys firm, behind-the-meter power in DFW.

Segment	Typical need	Why Sierra Solutions
GPU-cloud and AI-inference operators	10-20 MW, near-flat load, speed to market	Power in 18 months vs. a 4-5 year queue
Enterprise and healthcare data centers	Mission-critical uptime	99.99% firm power (N+2), islanded from grid events
Colocation operators expanding in DFW	Capacity for pre-leased halls	Predictable two-part tariff, 8.7 cents blended

- Examples expanding in DFW: CoreWeave and Lambda (Plano), Core Scientific (Denton), Aligned Data Centers.
- Fit: 10-20 MW (competition scope), 90%+ load factor, uptime-critical, 15-year contract.
- Credit tiers: investment-grade anchor supports 75% debt (base case); growth-stage tenant 65% debt plus a 12-month LC.
- Value: a ~10% power premium on revenue buys ~3.5 years of earlier revenue (see 'Is 8.7 cents competitive?').

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Customer Examples in DFW

Operators racing for AI capacity here today: the customer types we serve.

Company	Type	DFW activity (public announcements)
CoreWeave	AI cloud (GPU)	$1.6B AI cloud data center in Plano, operating since 2023
Lambda	AI cloud (GPU)	Taking capacity at Aligned's DFW-04 in Plano (2026)
Core Scientific	AI / HPC hosting	Denton campus hosting CoreWeave, ~260 MW critical IT
Aligned Data Centers	Colocation	DFW-04: ~$700M liquid-cooled AI facility in Plano
Hospital systems	Healthcare data centers	Mission-critical uptime; cannot tolerate curtailment

- Our fit: their 10-20 MW expansions and inference halls that cannot wait 5 years for the grid.
- These are examples of the customer type, not signed customers.

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Sierra Solutions: Business Model

Developer and operator: we originate, build, and run the asset.

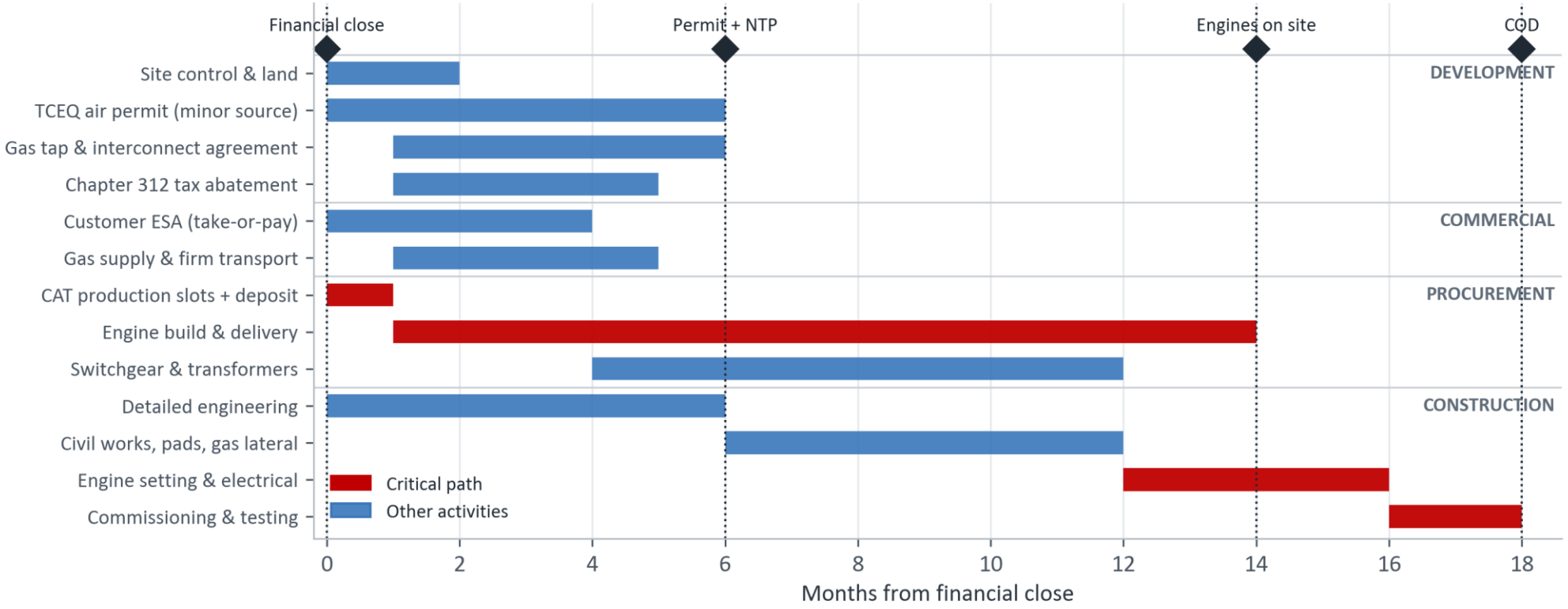
- Role: develop, finance, build and operate: permits, gas, customer contract, financing, construction, and 15 years of operation.
- Today: Sierra earns its development and construction-management fee (5% of hard costs, $1.34M), already inside the CAPEX.
- Project Dallas is our flagship: investors keep 100% of equity cash flows at the 19.98% base-case IRR. On this deal we build a track record, not our share.
- Next projects (Phase 2, other DFW sites): a standard developer promote, e.g. 20% of cash flows above a 15% investor IRR.
- For today's investors: first right to invest in Phase 2.

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Project Schedule: 18 Months to COD

Engine delivery is the critical path: production slots reserved at close.



Debt is committed at financial close; construction draws are released when the air permit is issued (month 6). The model conservatively charges interest from month 0.

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Phase 2: Expanding Red Oak to 31.5 MW

Shared site, gas tap and crew make the second block cheaper.

Metric	Phase 1	Phase 2 (incremental, indicative)
Firm capacity	15.75 MW (7 + 2 engines)	+15.75 MW (+8 engines; 17 total, N+3)
CAPEX	$31.0M ($1,970/kW firm)	$24.5M ($1,554/kW firm)
Build time	18 months	~12 months on an operating site
Firm availability	99.99%	99.998% combined (14 of 17 engines)
Blended tariff for Phase 1 returns	8.71 cents	7.21 cents

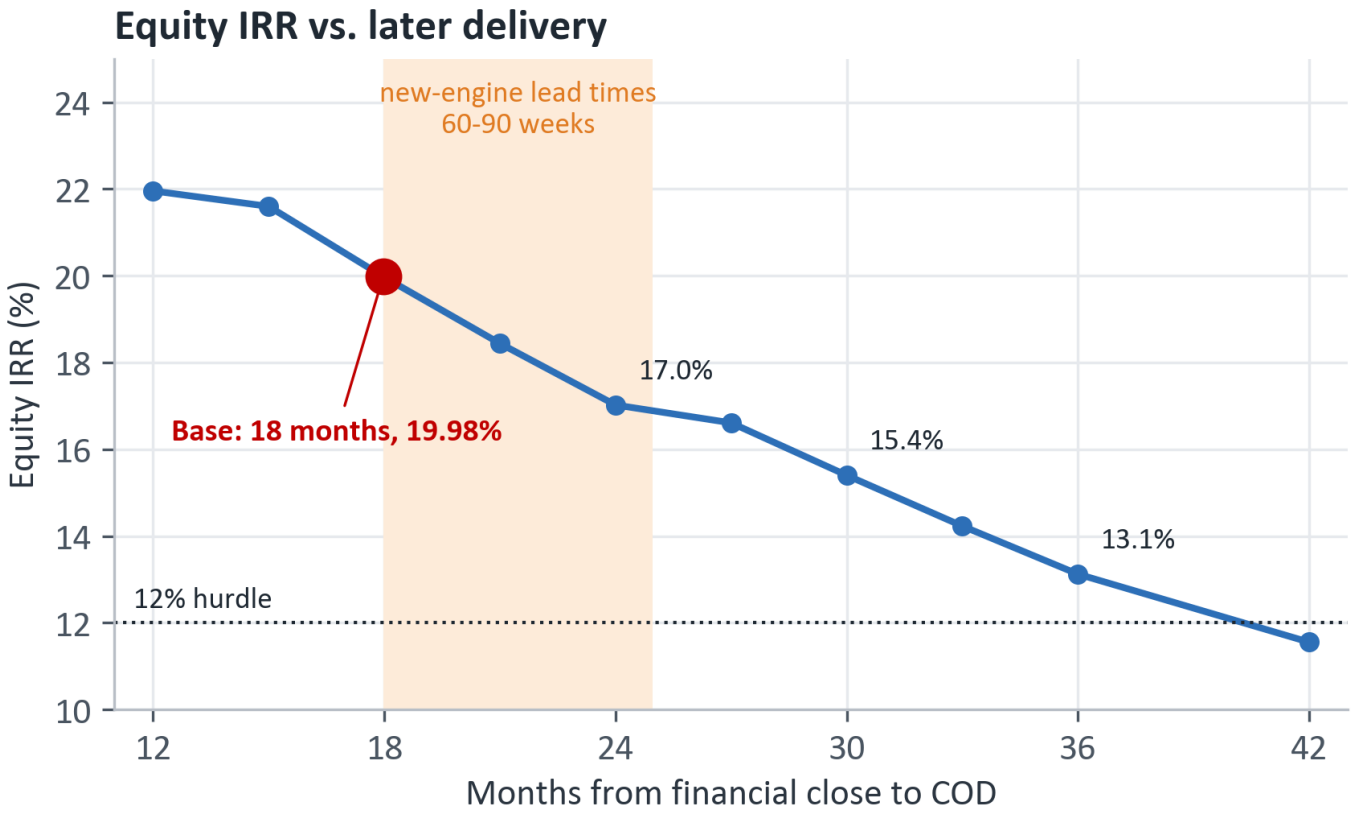
- Shared: gas tap and lateral, switchgear lineup, control room and crew (+50% staff); adds 3 acres of land.
- Shared spares: 17 engines carry 31.5 MW at 99.998%, one engine fewer than two separate N+2 plants.
- Pricing headroom: 7.21 cents blended still earns Phase 1's 19.98%.
- Constraints: tenant 2 must sit on the adjacent site; site NOx rises to ~21 tpy even with 90% SCR, just under DFW's 25 tpy major-source line; gas tap and firm transport must be upsized.

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Delivery Delay vs. Returns

Even a 6-month engine slip keeps equity returns at 17%.



- Engines are the critical path: each month of delay costs about 0.5 pts of IRR.
- At 24 months (top of today's 60-90 week lead times): 17.0% IRR, 1.44x min DSCR.
- The 12% hurdle holds until about 40 months: more than two years of slip.
- Delaying financial close until engine slots are confirmed costs little: interest only starts at close.

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Scaling Beyond 31.5 MW

The development template scales; the engine choice changes with size.

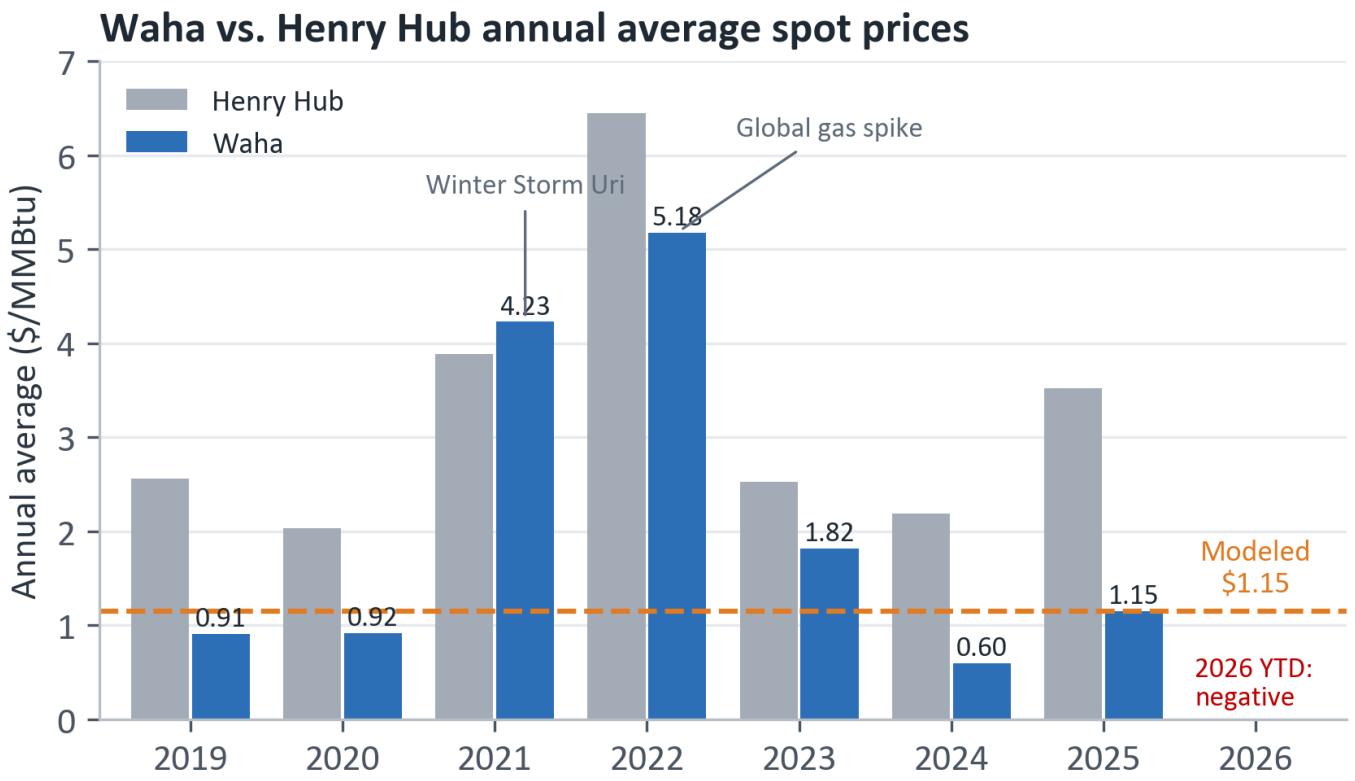
Scale	Best-fit generation	Why
10-30 MW (Project Dallas)	2-2.5 MW high-speed gas engines (CAT G3520H)	Fine-grained N+2, fast build, no cooling water
30-300 MW	Larger engine fleet, or 10-20 MW medium-speed engines	Fewer units and overhauls per MW, higher efficiency
300 MW+	GW-scale engine fleets, or engines + aeroderivative turbines	Already being built behind the meter for AI campuses

- Proven at scale: ~2 GW of Cat G3516 gensets ordered for the Monarch Compute Campus (WV); 613 MW of 20 MW gas engines permitted for Nexus Datacenter; a 200 MW behind-the-meter plant for Meta in Ohio.
- What changes with size: larger engines, a major-source air permit in DFW, and more firm gas capacity.
- Our answer: the same siting, gas, contracting and financing playbook; the engine fleet grows with the load.

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Gas Price Basis: Why $1.15 Waha

Built from 2019-2025 annual averages, not from a single year.



- 2019-2025 median $1.15; mean excluding Uri and the 2022 spike $1.08; 2024 $0.60; 2026 year-to-date negative.
- Annual volatility $1.83 implies ~$0.47 uncertainty on a 15-year average: our Monte Carlo spread.
- New takeaway (Hugh Brinson, Blackcomb, Q4 2026) may narrow the discount: hurdle still holds up to $2.10/MMBtu.
- Henry Hub $4.00 vs. 2025 average $3.52; it is only 5% of supply.

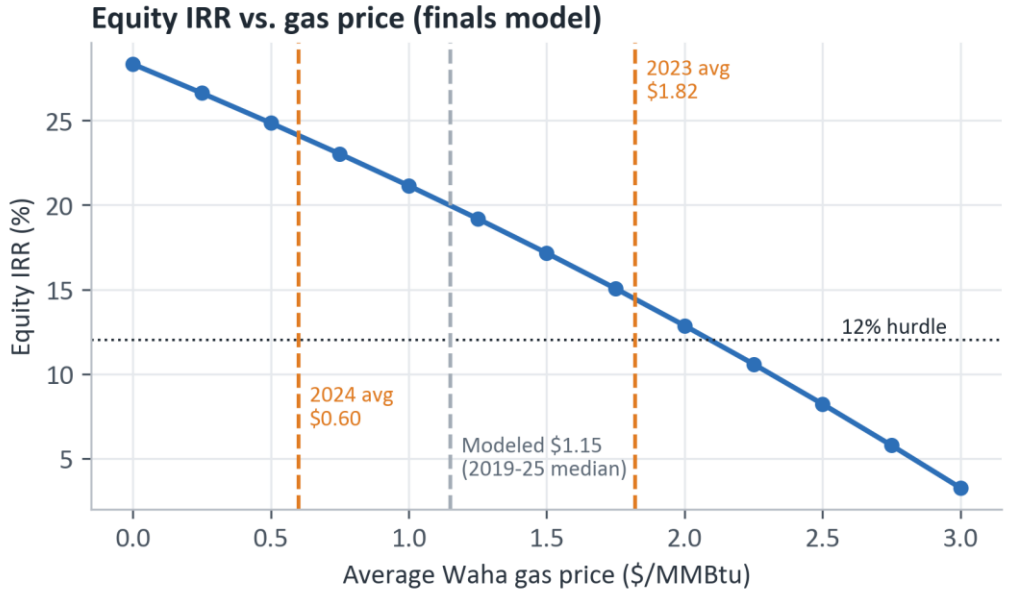
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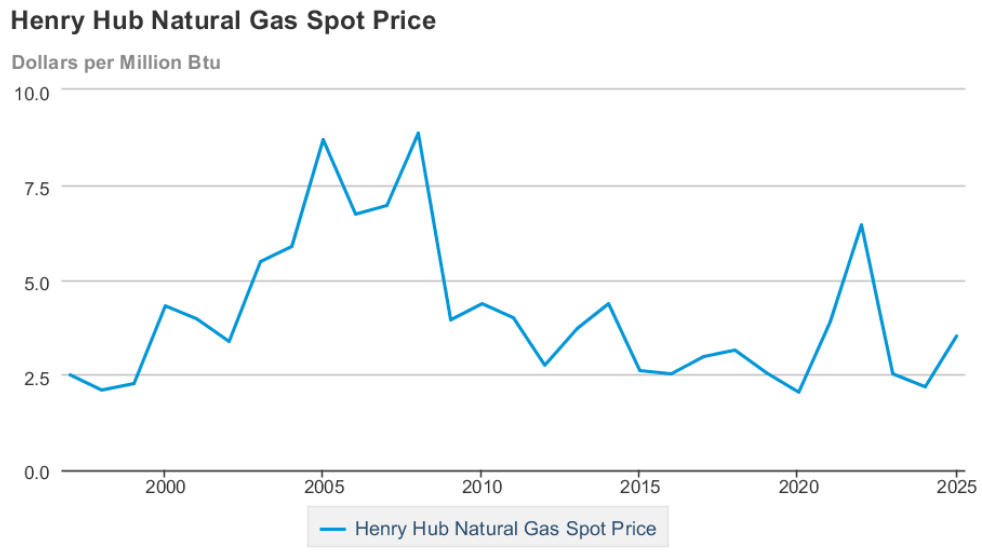
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Strategic Resilience: Fuel Conservatism

How equity returns move with the Waha price.

- Base case: Waha $1.15/MMBtu, the 2019-2025 median (and the 2025 average).
- At 2024's $0.60 average: 24.1% IRR. At 2023's $1.82: 14.4%.
- 12% hurdle holds up to $2.10/MMBtu; 1.25x DSCR holds up to $1.89/MMBtu.





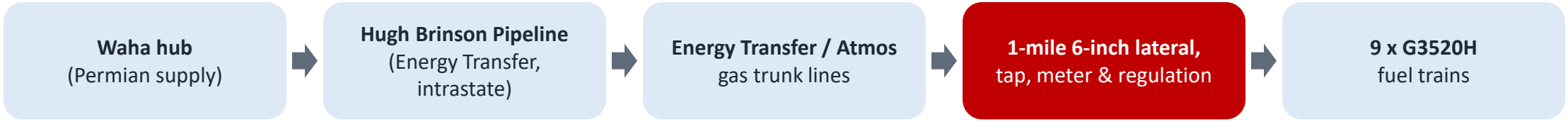
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Gas Supply Path and Interconnection

We tap existing trunk lines next to the site instead of building to Hugh Brinson.



- Why the trunk line: a 1-mile lateral (~$1M budgeted) instead of ~15 miles to Hugh Brinson; tapping an existing high-pressure system; two operators for supply diversity.
- Commercially: buy gas at the Waha index, firm transport on Hugh Brinson, delivered through Energy Transfer / Atmos at $0.60/MMBtu all-in.
- Physically, molecules arrive by displacement within the Energy Transfer network; what we contract is firm delivery capacity to our meter.
- Volume: ~3,262 MMBtu/d average, ~3,398 peak (about 3.3 MMcf/d), roughly 0.21% of Hugh Brinson's 1.5 Bcf/d first phase.
- On the site map, 'transmission lines' means gas transmission pipelines, not electric lines.

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Fuel Security

Winter Storm Uri was a supply-chain failure; here is how we are positioned.

- Firm, not interruptible, transport, across two operators' systems (Energy Transfer and Atmos).
- Post-Uri, Texas SB 3 (2021) requires weatherization across the gas supply chain and designation of critical gas facilities.
- On site: we specify CAT's cold-weather package and heated, enclosed fuel trains and gensets.
- N+2 engines plus dual-operator supply: no single on-site point of failure.
- Contract: regional gas curtailment is a defined force-majeure event in the ESA; availability damages are capped.
- A Uri-scale event hits every gas-fired asset in Texas; our design is no worse than the grid's own gas plants, and the customer avoids grid curtailment risk.

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Is 8.7 cents Competitive?

Priced above the grid average, and worth it for speed and uptime.

Benchmark	Price (cents/kWh)	Notes
Texas industrial average (EIA, 2025)	6.55	Comparable class for a 15.75 MW load
DFW data-center rate (CBRE, H1 2024)	~6.2	Our chart on slide 7
Texas commercial average (EIA, 2025)	8.64	Small and medium business
Oncor small commercial, all-in	8.9-9.2	Before demand charges
Sierra Solutions (behind the meter)	8.71	Firm, 99.99%, 18 months to power

- Premium vs. the industrial average: 2.2¢/kWh, about $2.9M/yr for the client.
- A ~12.1 MW-IT data center at CBRE's $196/kW-month (H2 2025) earns ~$28M/yr: the premium is ~10% of revenue.
- Waiting 3.5 years for the grid forgoes ~$100M of that revenue, plus diesel backup and curtailment risk.

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Pricing Flexibility: How Low Can We Go?

Negotiating room on the energy charge, capacity charge held at $22.50/kW-mo.

Energy rate	Blended rate	vs. 6.55 cents	Equity IRR	Min DSCR
5.50 cents (base)	**8.71¢**	**+2.16¢**	**19.98%**	**1.50×**
5.25¢	8.46¢	+1.91¢	17.71%	1.40×
5.00 cents (lender floor)	**8.21¢**	**+1.66¢**	**15.35%**	**1.31×**
4.75¢	7.96¢	+1.41¢	12.89%	1.21×
4.50¢	7.71¢	+1.16¢	10.33%	1.11×

- **About 0.5 cents of negotiating room without changing the capital structure.**
- Lender floor (1.30x min DSCR at 75% debt): blended 8.20¢, still 15.2% IRR.
- Hurdle floor (12% IRR): blended 7.87¢, but DSCR falls to 1.17×, which needs less debt (grey rows).
- Grid parity (6.55 cents) is below the cost of firm, redundant on-site gas power at this scale.

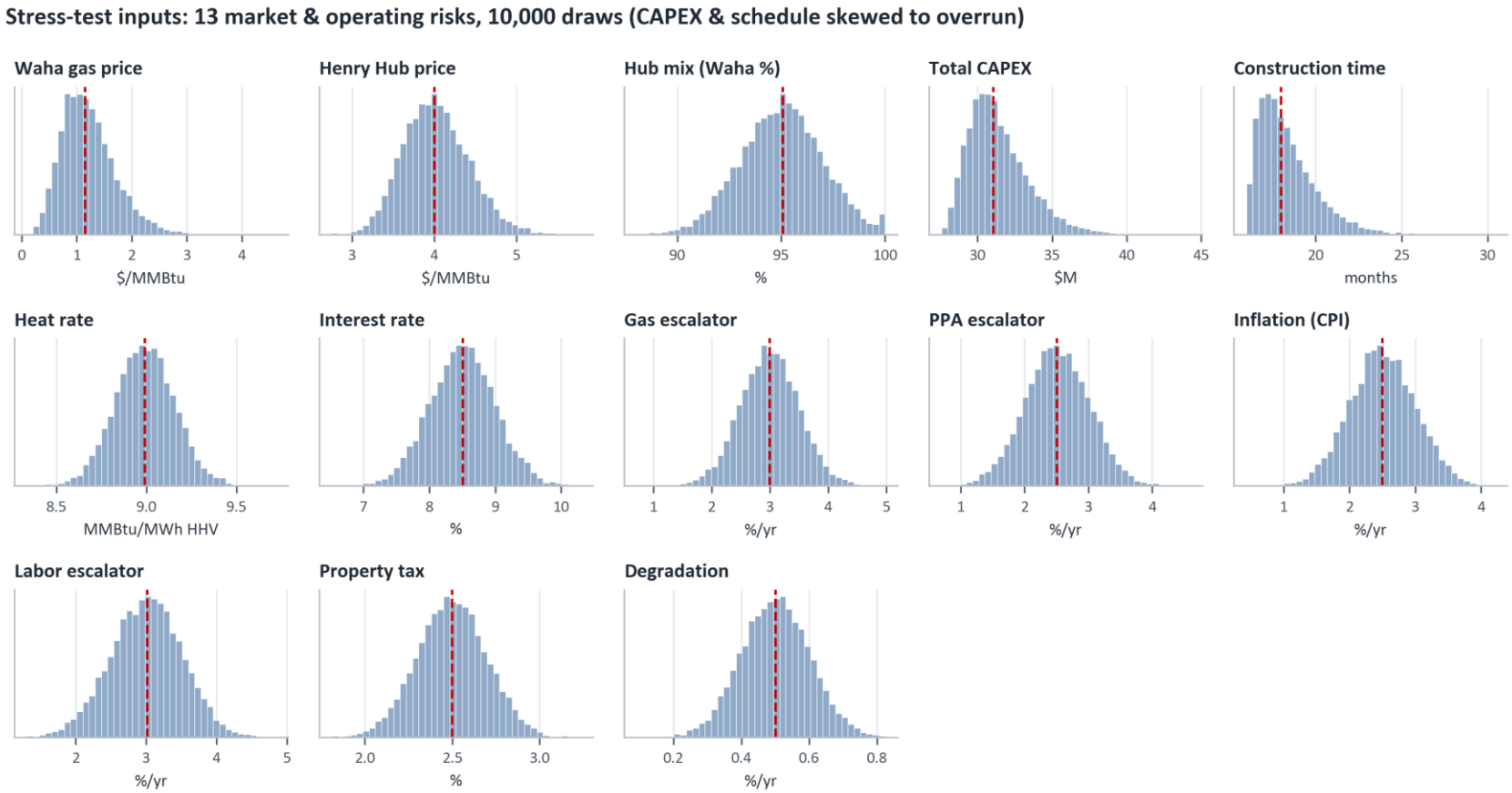
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Stochastic Boundary Conditions

Defining the Stress Portfolio for 10,000 Simulations

- Waha (15-yr average): median $1.15, right-skewed; sigma ~$0.47 = annual sigma $1.83 / sqrt(15).
- Henry Hub: median $4.00; sigma ~$0.40 = annual sigma $1.54 / sqrt(15).
- CAPEX: right-skewed, median = P50 estimate, overrun tail.
- Construction: median 18 months, slips up to ~6 months.
- Hub mix 90-100% Waha; interest +/-50 bp; escalators +/-0.5%.
- Energy & capacity rates are signed contract terms, not risks (see Pricing flexibility).

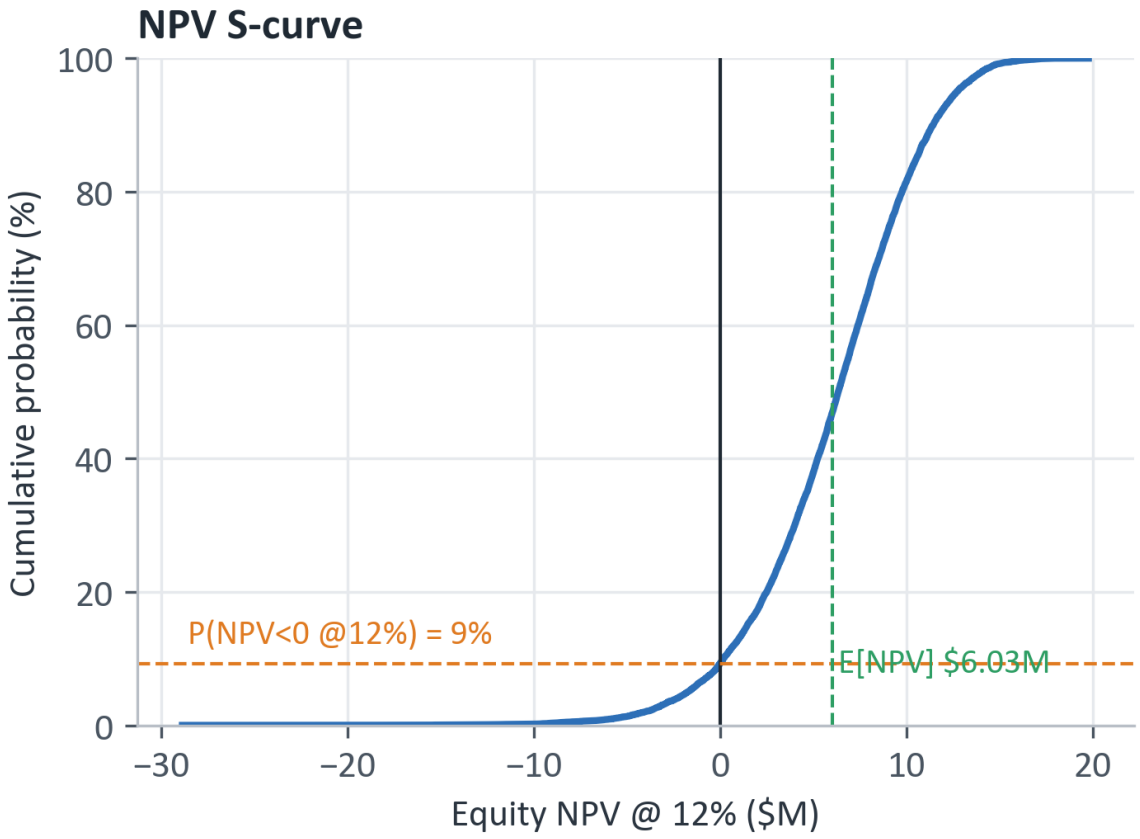


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Risk Metrics: NPV S-Curve

Decision metrics from 10,000 Monte Carlo runs.



- Expected NPV @ 12%: $6.03M (deterministic $6.92M)
- P(NPV < 0) = P(IRR < 12% hurdle): 9%
- P(capital loss, IRR < 0): 0.3%
- P(IRR < 8.5% cost of debt): 3%
- P(min DSCR < 1.25x lock-up): 14%
- P(min DSCR < 1.0x): 1.7%, bridged by the 6-month DSRA
- Median MOIC: 5.3×

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Risk Register

Every major risk, its mitigation, and what remains.

Risk	Mitigation	Residual
Waha discount narrows	Base at 7-yr median $1.15; hurdle holds to $2.10/MMBtu; skewed Monte Carlo	Medium
Offtaker credit	15-yr take-or-pay; parent guarantee or 12-month LC; relocatable gensets	Medium
Engine lead time	Early CAT slot reservation; surplus-market sourcing already priced in CAPEX	Medium
Construction delay / overrun	Fixed-price, date-certain EPC with delay LDs; 10% contingency	Low-Med
Fuel interruption	Firm transport, two operators, SB 3 weatherization	Low-Med
Engine failure	N+2 (99.99%), CAT service agreement, $250k critical spares	Low
Interest rate	Swap at financial close; +/-50 bp in Monte Carlo	Low
Air permit	0.5 g NOx engines plus SCR for TCEQ compliance	Low
Tax classification	15-yr MACRS case still 19.38% IRR	Low

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Offtaker Credit and Asset Redeployability

What happens if the tenant fails?

- Take-or-pay capacity payments alone cover 125% of debt service.
- Credit support: parent guarantee, or a 12-month letter of credit of about $4.3M (one year of capacity charges).
- Skid-mounted G3520H gensets are relocatable, and our CAPEX already prices units from an active secondary market.
- Re-tenanting: North American data-center vacancy is at record lows (CBRE, 2025), and DFW is a top market.
- The 6-month DSRA (~$1.81M) bridges the re-marketing period.

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Project CAPEX: Line-Item Detail

Detailed breakdown of hard and soft infrastructure costs indicates industrial average.

	Item	Qty	Unit	Unit Rate ($)	Total Cost ($)	Source / Logic
Power Generation Modules	CAT G3520H Generator Set (Skid)	9	Each	$1,725,000	$15,525,000	Market Quote (Surplus Record, 2025 Avg).
	Sound Attenuated Enclosure (ISO)	9	Each	$175,000	$1,575,000	MachineryTrader listing for 40' container w/ 85dBA rating.
	SCR Emissions Package	9	Each	$225,000	$2,025,000	EPA Retrofit Data ($100/kW) for NOx compliance.
Electrical BoP	15kV Switchgear Sections	12	Cell	$75,000	$900,000	Siemens GM-SG Style Vacuum Breaker (Installed New).
	Master Control Panel (Sync)	1	Lot	$650,000	$650,000	Complex Logic for 9-Unit Paralleling (Est. based on CAT EMCP 4.4).
	MV Cabling & Trays	3000	Ft	$150	$450,000	15kV Shielded Cable + Tray Install ($150/ft installed).
	Step-Up Transformers (2.5 MVA)	5	Each	$120,000	$600,000	Padmount, 13.8kV/480V. One per 2 engines + spare.
	Grid Connection Hardware	1	Lot	$400,000	$400,000	Overhead line + disconnect switch to Data Center
Mechanical BoP	Gas Fuel Train (Regulators)	9	Each	$35,000	$315,000	High-pressure regulators per engine inlet.
	Lube Oil Storage System	1	Lot	$150,000	$150,000	Central 5,000 gal tank + piping loop.
	Gas Interconnection Lateral	1	Mile	$1,000,000	$1,000,000	6" Lateral to Atmos/ET Trunk Line.
Civil & Site Works	Concrete Equipment Pads	9	Each	$25,000	$225,000	15 cy reinforced concrete per pad @ $1,500/cy.
	Site Grading & Fencing	4	Acre	$75,000	$300,000	Heavy grading + Anti-climb security fence.
	Land Acquisition	4	Acre	$250,000	$1,000,000	Red Oak Industrial Park Listing (2025).
	Spare Parts	1	Lot	$250,000	$250,000	Critical spares for Day 1 reliability.
Soft Costs	Engineering & Permitting	1	Lot	$1,500,000	$1,500,000	Air Permit (TCEQ) + Elect. Interconnect Study.
	Construction Management	1	Lot	5%	$1,343,250	5% of Hard Costs.
	Contingency (P50)	1	Lot	10%	$2,820,825	10% of hard costs (P50 project-finance reserve).
	TOTAL PROJECT COST				**$31,029,075**	
	$/kW (Installed)				**$1,532**	

- Includes $2.8M P50 contingency and $1.5M permitting/engineering.
- $1,532/kW installed ($1,970/kW firm), within NREL's $900-$2,600/kW range [1].

[1] nrel.gov

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Heat Rate Basis: LHV vs. HHV

CAT quotes LHV; gas is bought on HHV. We model what we pay for.

G3520H configuration (CAT LEXE1552-02)	Load	LHV Btu/kWh	HHV Btu/kWh (x1.108)	Efficiency (HHV)
Humidity/fuel tolerant, 0.5 g NOx (modeled)	100%	8,114	8,990	38.0%
Humidity/fuel tolerant, 0.5 g NOx	75%	8,305	9,202	37.1%
Humidity/fuel tolerant, 0.5 g NOx	50%	8,781	9,729	35.1%
Humidity/fuel tolerant, 1.0 g NOx + our SCR	100%	7,855	8,703	39.2%

- Modeled: 8,990 Btu/kWh HHV + 0.5%/yr degradation (regional model used the LHV figure: -1.8 pts IRR when corrected).
- Upside: our SCR lets the engines run CAT's 1.0 g/bhp-hr tune and still meet TCEQ limits: 8,703 Btu/kWh HHV, +0.5 pts IRR.
- If all 9 engines spin at ~75% load for transient cover: 9,202 Btu/kWh HHV, -0.3 pts IRR.
- Gas demand: ~3,262 MMBtu/d average, ~3,398 MMBtu/d peak (about 3.3 MMcf/d).

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N+2 Reliability Math

P(at least 7 engines = 15.75 MW firm available), independent forced outages.

Per-engine availability	N (7 units)	N+1 (8 units)	N+2 (9 units)
97%	80.80% (1,682.1 h/yr)	97.77% (195.7 h/yr)	99.80% (17.3 h/yr)
98%	86.81% (1,155.2 h/yr)	98.97% (90.6 h/yr)	99.94% (5.4 h/yr)
99%	93.21% (595.1 h/yr)	99.73% (23.6 h/yr)	**99.992% (42 min/yr)**

- At a 1% forced-outage rate per engine, N+2 delivers 99.99% firm availability (~42 min/yr).
- N+1 would leave ~24 h/yr exposed; N (no redundancy) fails ~25 days/yr.
- Planned maintenance is scheduled one engine at a time, so N+1 cover is kept during service.
- Common-mode risks (gas supply, switchgear) addressed by dual-operator gas supply and 12-cell sectionalized 15 kV switchgear.

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Technology Selection Detail

Modular engines outperforming turbines for high-reliability 15MW loads.

Name	Price ($M/unit)	N Configuration Units (N)	N Configuration Total CapEx ($M)	N+1 Configuration Units (N+1)	N+1 Configuration Total CapEx ($M)	N+2 Configuration Units (N+2)	N+2 Configuration Total CapEx ($M)	2N Configuration Units (2N)	2N Configuration Total CapEx ($M)
Titan 130	10.925	2	21.85	3	**32.78**	4	43.70	4	43.70
W31SG	17.655	2	35.31	3	**52.97**	4	70.62	4	70.62
G3520H	1.725	7	12.08	8	13.80	**9**	**15.53**	14	24.15

Name	Net Efficiency	Fuel Burn (MMBtu/hr)	Δ Fuel (MMBtu/hr)	Annual Fuel Penalty ($M/yr)	Project Fuel Penalty ($M)	Power, Net (MW)
Titan 130	0.354	144.58	45.39	0.60	8.95	11.92
W31SG	0.516	99.19	0.00	0.00	0.00	10.26
G3520H	0.421	121.57	22.38	0.29	4.41	2.25



westernstatescat.com

- 9 modular CAT G3520H units.
- N+2 Configuration: 15.75 MW Firm / 20.25 MW Installed.
- **Physics-based modeling:** 8,114 Btu/kWh (LHV) / 8,990 (HHV) heat rate with 0.5% annual mechanical degradation.

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Competitive Analysis: Technology Selection Data

Comparative CAPEX analysis driving the decision for Modular Reciprocating Engines.

- Solar Titan 130 Turbine pricing (10.925M per unit) sourced from distributed generation market reports.
- Wartsila 31SG pricing (17.655M perunit) derived from techno-economic literature on flexible gas power plants.
- Granularity: losing one G3520H removes 2.25 MW (14% of load); losing one Titan 130 removes 11.9 MW (76%). Small blocks also follow AI inference load steps.

Emissions, Water and Noise

Permit-ready air, near-zero water.

~10 tpy NOx
with 90% SCR: a minor source, under DFW's 25 tpy major-source line

~0 process water
radiator cooling; a combined-cycle plant would use ~27M gal/yr

0.48 tCO2/MWh
about half the intensity of coal power

- Air: Red Oak is in the DFW ozone nonattainment area (Ellis County, 'severe'), where 25 tpy of NOx defines a major source. Without SCR we would emit ~104 tpy; the $2.0M SCR keeps us a minor source and the permit on a ~6-month track.
- Water: radiator-cooled engines need no cooling towers. A combined-cycle plant of the same output would consume ~205 gal/MWh (NREL median), about 27 million gallons a year.
- Carbon: ~63,183 tCO2/yr (EPA 53.06 kg/MMBtu on HHV).
- Noise: enclosures rated to 85 dBA. Future upside: exhaust heat could drive absorption chilling for the data center (not modeled).

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Commercial Structure: Key Agreements

Proposed term sheet behind the cash flows.

Agreement	Key terms (as modeled)
Gas supply	15-yr, Waha-index priced (95%) with Henry Hub-linked balance (5%); ~3,262 MMBtu/d average, ~3,398 peak.
Firm transport	Hugh Brinson Pipeline into Energy Transfer / Atmos trunk lines at $0.60/MMBtu firm; project builds 1-mile 6-inch lateral.
Power offtake (ESA)	15-yr take-or-pay: $22.50/kW-mo on 15.75 MW (+2.5%/yr) + $0.055/kWh energy (+3%/yr); 99.99% availability guarantee with LDs; parent guarantee or 12-month LC.
O&M / LTSA	CAT dealer service agreement incl. overhauls at $17.50/MWh (+CPI); on-site staff $0.92M/yr (+3%).
EPC	Fixed-price, date-certain with delay LDs; 10% P50 contingency ($2.8M).
Senior debt	75% non-recourse, 8.5%, 13-yr amortizing (2-yr tail inside the ESA); 6-month DSRA; 1.20x lock-up / 1.05x default.
Tax & incentives	MACRS 5-yr with loss carryforward; Dallas County Chapter 312 abatement structure on property tax.

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Data Sources: Financial & Commercial Inputs

Benchmarking cost-basis and revenue assumptions against 2025 market data.

- CAT G3520H Genset Price ($1,725,000): Sourced from Surplus Record (2025 Avg) market quotes.
- Sound Enclosure ($175,000): Sourced from MachineryTrader listings for 85dBA rated ISO containers
- SCR Emissions Package ($225,000): Sourced from EPA Retrofit Data (approx. $100/kW for NOx compliance).
- 15kV Switchgear ($75,000/cell): Pricing based on Siemens GM-SG Style Vacuum Breakers.
- Land Acquisition ($250,000/acre): Sourced from Red Oak Industrial Park real estate listings (2025).
- MV Cabling ($150/ft): Industry estimate for 15kV shielded cable + tray installation.
- Concrete/Civil Works: Unit rates ($1,500/cy) based on standard civil engineering estimates.
- Electricity Price Benchmarks ($0.06 - $0.09/kWh) sourced from CBRE Data Center Trends reports.
- Hurdle (12%), DSCR (1.25x), LLCR (1.4x), MOIC and debt-share ranges from SPE briefing (Stephen Janssen); risk metrics per SPE briefing (Greg Terzian).
- Tax Depreciation (MACRS 5-Year) based on IRS Publication 946 for eligible power generation property.
- Gas price basis: Waha annual averages 2019-2025 (EIA and industry; median $1.15); Henry Hub 2025 average $3.52 (EIA). Texas industrial power 6.55 cents/kWh (EIA 2025).

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Data Sources: Technical & Operational Inputs

Physics-based modeling parameters anchored in OEM specifications and federal data.

- Engine Performance: 8,114 Btu/kWh LHV = 8,990 Btu/kWh HHV (x1.108), Caterpillar G3520H spec sheet LEXE1552-02.
- Interconnection Queue delays (60+ Months) sourced from Interconnection.fyi and LBNL Empiricist Reports.
- Grid Curtailment trends sourced from S&P Global Market Intelligence and EIA.gov.
- Natural Gas Spot Prices (Waha and Henry Hub) sourced from EIA Historical Data Series.
- Net-to-ISO Derating Factor (0.90) applied to account for Dallas ambient temperatures (105°F) and parasitic loads.
- Annual Mechanical Degradation (0.5%) based on standard reciprocating engine lifecycle curves.
- Firm Transport Fees ($0.60/MMBtu) based on standard intrastate Texas pipeline tariffs.
- Property Tax Abatement structure modeled on Dallas County Chapter 312 economic incentives.

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Engine Room

Validating investment integrity through a synced Python-Excel framework.

- **Excel Architecture:** Serving as the static "Ground Truth" for all engineering, commercial, and financial inputs to ensure auditability.
- **Python Integration:** Custom "Digital Twin" script reads Excel inputs dynamically to perform 15-year cash flow waterfall calculations.
- **Stochastic Layer:** Python engine executes 10,000 Monte Carlo iterations on 13 market & operating risks around the Excel base case.

Project Inputs		Finance Model								MACRS Table		
Model Start Date	1-Jan-27	**Investment**		**Fuel Supply**		**Revenue & OpEx**		**Power Generation**		**MACRS Year**	**MACRS Rate (%)**	
Construction period (years)	1.5	Debt (fraction)	0.75	Gas Price Waha ($/MMBtu)	1.15	Capacity Rate ($/kW-mo)	$22.50	Active # of Turbines	7	1	20	
Operation period (years)	15	Equity (fraction)	0.25	Gas Price Henry ($/MMBtu)	4	Energy Rate ($/kWh)	$0.055	Total # of Turbines	9	2	32	
Delivered Power (MW)	15.75	Loan Interest Rate (%)	8.5	Gas Fraction Waha	0.95	Income Tax (%)	21	Unit Price ($)	$1,725,000.00	3	19.2	
		DSRA ($)	$1,810,813	Gas Fraction Henry	0.05	Base Insurance (%)	0.75	Unit Net Power (MW)	2.25	4	11.52	
		Hard Capex ($)	$31,029,075.00	Avg Gas Price ($/MMBtu)	1.292 5	Base Admin ($)	$75,000.00	Peak power (MW)	20.25	5	11.52	
		Initial Loan	$24,629,916	Gas Transport Fee ($/MMBtu)	0.6	Variable O&M Base (%)	17.5	Size (acres)	4	6	5.76	
		Property				Staffing Base ($)	$923,000.00	Capacity Factor	0.96	7	0	
		Land Price (Acre/MW)	0.2			**Escalators**		Heat Rate (Btu/MWh)	8.114			
		Property Tax (%)	2.5			Inflation (CPI) (%)	2.5	HHV/LHV (%)	10.8			
		Tax Drop (%)	5			PPA Escalator (%)	2.5	Degradation Factor (%)	0.5			
						Wage Escalator (%)	3					
						Gas Escalator (%)	3					

```python
def run_dallas_twin_model(p):
    # Sourcing Strategy
    weighted_hub_price = (0.95 * p['waha_gas_base']) + (0.05 * p['henry_gas_base'])
    initial_loan_draw = (p['capex_hard_soft'] + p['dsra']) * p['debt_share']
    initial_equity_y0 = (p['capex_hard_soft'] + p['dsra']) * (1 - p['debt_share'])

    # MACRS 5-Year Table (20%, 32%, 19.2%, 11.52%, 11.52%, 5.76%)
    macrs_table = [0, 0.20, 0.32, 0.192, 0.1152, 0.1152, 0.0576] + [0]*40
    first_full_ops_year = int(np.ceil(p['const_years'])) + 1

    results = []
    curr_loan_bal = initial_loan_draw
    balloon_bal = 0

    for yr in range(p['total_range']):
        # Construction / operations switching
        rf = max(0, min(1, yr - p['const_years'])) if yr > 0 else 0
        ti = (max(yr-1, p['const_years']) + yr)/2 if rf > 0 else 0

        # Revenue
        total_rev = ((p['gen_mwh'] * 1000 * p['en_rate_base'] * ((1 +
p['gas_esc'])**ti)) + \
                     (p['mw_delivered'] * 1000 * p['cap_rate_base'] * 12 * ((1 +
p['ppa_esc'])**ti))) * rf

        # Opex
        fuel = p['gen_mwh'] * (p['hr'] * ((1 + p['degrad_rate'])**ti)) * \
               ((weighted_hub_price * ((1 + p['gas_esc'])**ti)) +
p['transport_fee']) * rf
        var_om = (p['gen_mwh'] * 17.50 * ((1 + p['cpi_esc'])**ti)) * rf
        staff = (p['staff_base'] * ((1 + p['labor_esc'])**ti)) * rf

        stat_on = 1 if yr > 0 else 0
        ins_adm = (p['ins_base'] + p['adm_base']) * ((1 + p['cpi_esc'])**ti) if rf >
0 else (p['ins_base'] + p['adm_base']) * stat_on
        tax_p = (p['capex_hard_soft'] * p['prop_tax']) * ((1 -
p['tax_drop'])**max(0, yr - 2)) * stat_on
        ebitda = total_rev - (fuel + var_om + staff + ins_adm + tax_p)

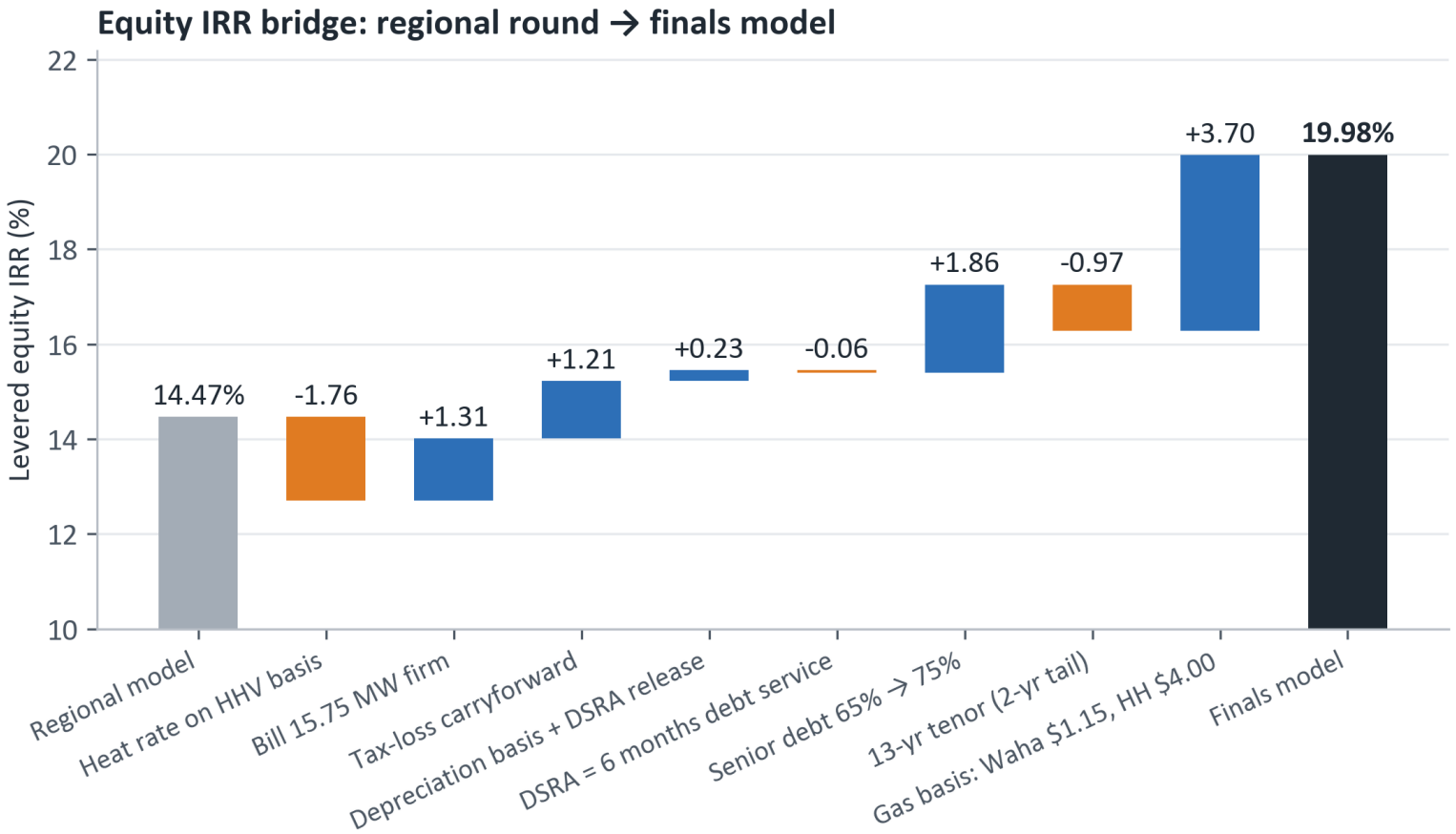
        # Debt mechanics
        interest = (curr_loan_bal * p['interest_rate']) if yr > 0 else 0
        idc = interest * (1 - rf) if yr > 0 else 0
```

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Model Refinement

Refinements since the regional round, incorporating judges' feedback.



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